13.3.1 RTO and RPO comparison for a single EC2 instance

You've learned about two possible solutions for making a single EC2 instance highly available. When choosing the solution, you have to know the application's business requirements. Can you tolerate the risk of being unavailable if an availability zone goes down? If so, EC2 instance recovery is the simplest solution, where you don't lose any data. If your application needs to survive an unlikely availability zone outage, your safest bet is autoscaling with data stored on EFS, but this method also has performance effects compared to storing data on EBS volumes. As you can see, there is no one-size-fits-all solution. You have to pick the solution that fits your business problem best. Table 13.2 compares the solutions.

Table 13.2 Comparison of high availability for a single EC2 instance

	RTO	RPO	Availability
EC2 instance, data stored on EBS root volume: recovery triggered by a CloudWatch alarm	About 10 minutes	No data loss	Recovers from a failure of a virtual machine but not from an outage of an entire availability zone
EC2 instance, data stored on EBS root volume: recovery triggered by autoscaling	About 10 minutes	All data lost	Recovers from a failure of a virtual machine and from an outage of an entire availability zone
EC2 instance, data stored on EBS root volume with regular snapshots: recovery triggered by autoscaling	About 10 minutes	Realistic time span for snapshots: between 30 minutes and 24 hours	Recovers from a failure of a virtual machine and from an outage of an entire availability zone
EC2 instance, data stored on EFS filesystem: recovery triggered by autoscaling	About 10 minutes	No data loss	Recovers from a failure of a virtual machine and from an outage of an entire availability zone

If you want to be able to recover from an outage of an availability zone and need to decrease the RPO, you should try to achieve a stateless server. Using